

Calibration of the Quantitative Model

Provisional update

Tommaso De Santo

1 Calibration

I organize the model parameters into two groups. The first consists of parameters assigned from external evidence, measured directly in the data, or fixed as maintained restrictions. The second consists of parameters estimated internally by matching model moments to their empirical counterparts. I describe the two groups in turn, explain which moments are particularly informative about the internally estimated parameters, and then report the model's fit.

1.1 Parameters Set Outside the Estimation

I first describe the parameters assigned outside the estimation. Table 1 summarizes their values and sources.

Demographics and preferences. A model period is four years. Households enter adult life at age 18, retire at age 66, and live at most through age 85. Survival is certain before retirement and thereafter follows four-year probabilities constructed from the combined male and female survivor counts in the 2023 Social Security period life table. Dependent children remain in the household for an expected 18 years, implying a maturity probability of $2/9$ per period. I set risk aversion to $\sigma = 2$.

Housing finance and transaction costs. Following Greaney et al. (2025), I set the annual real return on liquid wealth to $i_{\text{ann}} = 0.02$, annual housing depreciation to $\delta_{\text{ann}} = 0.011$, the proportional selling cost to $\psi^s = 0.06$, and the labor-income tax rate to $\tau_{\text{pay}} = 0.179$. The return is based on the average ten-year real interest rate, depreciation combines BEA residential-structure depreciation with an adjustment for the land share of house values, and the tax rate comes from OECD data. I set the annual property-tax rate to $\tau_{\text{ann}}^p = 0.01$. Mortgage debt cannot exceed a share $\phi = 0.80$ of the value of the house, so a purchase requires at least 20 percent equity. The benchmark imposes no separate payment-to-income limit.

Income process. I impose a deterministic lifecycle profile of household earnings, e_a , normalized to average one over working ages. Within age, log income follows an annual AR(1) process with persistence $\rho_z = 0.960$, which I sample at the model's four-year frequency and approximate with a five-state Rouwenhorst chain. This persistence is near the upper end of the PSID-based estimates summarized by Sommer and Sullivan (2018). I set the annual innovation standard deviation to $\sigma_z = 0.0645$, below estimates of gross income risk in that literature. I treat this variance as provisional rather than as an externally validated estimate of income risk.¹ The quantitative results are conditional on this restriction.

¹With innovation risk in the empirical range, households in the lowest persistent-income states cannot always finance the model's hard Stone–Geary minimum bundle. The current model omits insurance margins that could resolve this feasibility conflict.

Housing products and supply. Housing is indivisible and available in a limited assortment of types and sizes (Davis and Van Nieuwerburgh, 2015). I represent this lumpiness with five owner-occupied house sizes, $\{H_k\}_{k=1}^5 = \{2, 4, 6, 8, 10\}$ rooms. Renters choose housing continuously up to $\bar{h}^R = 6$ rooms, the 90th percentile of the rental-room distribution in the matched ACS sample. These menus summarize the observed tenure-specific distribution of rooms. I set the housing-supply elasticity to $\eta = 1.75$, following the population-weighted metropolitan estimate in Saiz (2010).

Entry wealth and bequests. Households enter at age 18 with a draw from the PSID net-worth-to-income distribution of childless renters aged 18–24. I impose $\theta_n = 0$, so bequest utility does not vary directly with the number of children. Children can still affect estate accumulation through their effects on housing needs, tenure, and saving.

Table 1: Parameters set outside the estimation

Parameter	Value	Source or restriction
Model period	4 years	Model timing
Entry, retirement, and maximum ages	18, 66, and 85	Model timing
Post-retirement survival	Age-specific	2023 Social Security period life table
Expected years with dependent children	18 years	Child-maturity restriction
Risk aversion σ	2	Maintained curvature
Annual real return i_{ann}	2.0%	Average 10-year real rate, 1962–2024; Greaney et al. (2025)
Annual housing depreciation δ_{ann}	1.1%	BEA structures depreciation, adjusted for the residential land share; Greaney et al. (2025)
Annual property tax τ_{ann}^p	1.0%	Maintained tax rate
Financed share ϕ	0.80	20% down-payment restriction; Davis and Van Nieuwerburgh (2015)
Sale cost ψ^s	6%	Davis and Van Nieuwerburgh (2015); Greaney et al. (2025)
Labor-income tax τ_{pay}	17.9%	OECD U.S. average; Greaney et al. (2025)
Income persistence ρ_z	0.960 annually	See text
Income innovation standard deviation σ_z	0.0645 annually	See text
Owner house sizes $\{H_k\}$	$\{2, 4, 6, 8, 10\}$ rooms	Occupied-room distribution in the ACS; see text
Largest rental unit $\bar{h}^R$	6 rooms	Matched ACS rental-size distribution
Housing-supply elasticity η	1.75	Saiz (2010)
Entrant wealth distribution	Wealth-to-income distribution	Childless renters aged 18–24 in the PSID
Child shifter in bequest utility θ_n	0	Maintained restriction

1.2 Internally Estimated Parameters

I estimate the remaining 14 parameters using 14 moments describing saving and bequests, baseline consumption and housing demand, child costs and family size, tenure, and aggregate housing. The parameters are chosen jointly. The mapping is nevertheless structured in low-dimensional blocks: a three-parameter saving–bequest block, a three-parameter baseline demand block, a two-parameter fertility block, and six parameters with distinct primary counterparts. Table 2 reports the estimates, search regions, and the primary empirical counterpart of each parameter.

Table 2: Internally estimated parameters and primary empirical counterparts

Parameter	Estimate	Search region	Near bound	Primary empirical counterpart
β_{annual}	0.9991	[0.800, 0.9995]	Yes	Aggregate wealth / annual gross labor earnings
θ_0	0.0460	[0, 8]	Yes	Annual bequest flow / aggregate wealth
θ_1	0.0944	[0.02, 16]	Yes	Late-life total-wealth p90/p50
α_c	0.6145	[0.02, 0.98]	No	Childless-renter rent-expenditure slope
$\bar{c}_0$	0.5090	[0, 2]	No	Childless-renter intercept at the mean rent per room
$\bar{h}_0$	0.5139	[0.05, 5.8]	No	Mean rooms among bottom-quintile childless renters
$\bar{c}_n$	0.1553	[0, 3]	No	Family-size coefficient in the consumption auxiliary
$\bar{h}_{\text{jump}}$	0.0522	[0, 8]	Yes	Housing increment associated with the first child
$\bar{h}_n$	0.9258	[0, 5]	No	Rooms gap: 3+ versus 1–2 children
ψ_{child}	−1.0261	[−3, 3]	No	Completed fertility, jointly with childlessness
κ_F	5.4615	[0.02, 50]	No	Childlessness, jointly with completed fertility
χ_O	1.0745	[0.1, 5]	No	Ownership rate, ages 30–55
κ_T	0.0190	[0, 0.12]	No	Four-year residual tenure Brier score
$\bar{H}$	4.9399	[0.2, 80]	No	Mean occupied rooms, ages 18–85

Saving. The annual discount factor governs households’ willingness to carry resources across ages. Its primary target is aggregate household net worth divided by annual gross labor earnings, measured in the 2005–2019 PSID. The model constructs the same ratio: beginning-of-period liquid wealth plus gross housing value in the numerator, and gross labor earnings of working households in the denominator. Mortgage debt is already contained in liquid wealth. The target is 6.873; this measured gross/gross ratio replaces the previously borrowed 6.9 ratio with an after-tax denominator. Age-binned wealth ratios remain untargeted lifecycle diagnostics.

Housing demand, tenure, and supply. For a childless renter with allocatable expenditure $x = c + rh$, the Stone–Geary first-order condition implies

$$rh = -(1 - \alpha_c)\bar{c}_0 + (1 - \alpha_c)x + \alpha_c r \bar{h}_0. \quad (1)$$

This creates a triangular three-parameter block. The rent-expenditure slope is most informative about α_c ; the intercept at the sample’s mean rent per room combines $\bar{c}_0$ and $\bar{h}_0$; and mean rooms among childless renters in the bottom expenditure quintile pins down the housing level and separates the two minimum requirements. The same sample restrictions, trimming rule, and auxiliary regressions are applied to the CEX and simulated model data. The parameters are therefore estimated jointly from three moments; none of the three empirical coefficients is itself a structural parameter.

The owner-service premium χ_O is primarily disciplined by the ownership rate at ages 30–55. The tenure dispersion κ_T is disciplined by a four-year Brier score: the mean squared error in predicting ownership four years ahead using only state variables that can be constructed in both the PSID and the model. This auxiliary moment isolates residual tenure variation more directly than the unconditional switching rate. Finally, the supply scale $\bar{H}$ is disciplined by mean occupied rooms at ages 18–85.

Consumption needs and family size. The consumption requirement is $\bar{c}_0 + \bar{c}_n n_s$. The empirical counterpart for $\bar{c}_n$ is the coefficient on family size in a consumption regression that also controls for rooms, income, age, and ownership; the identical regression is run on the simulated population. Thus the target 0.0752 is an auxiliary coefficient that disciplines $\bar{c}_n$, not the numerical value of $\bar{c}_n$ itself. The housing requirement is $\bar{h}_0 + \bar{h}_{\text{jump}} \mathbf{1}\{n_s \geq 1\} + \bar{h}_n n_s$. The first-child rooms increment is most informative about $\bar{h}_{\text{jump}}$, while the rooms difference between households with three or more children and those with one or two is most informative about $\bar{h}_n$. These are also model–data moment comparisons, rather than identities between reduced-form coefficients and primitives.

Completed fertility and childlessness form a two-parameter, two-moment block for $(\psi_{\text{child}}, \kappa_F)$. The average utility value of children primarily shifts the level of completed fertility, while the choice-dispersion parameter primarily governs how much mass remains childless. Because both parameters move both moments, the pair is estimated jointly.

Bequests. Bequeathable wealth is liquid wealth plus gross owner-occupied housing. The child-blind warm-glow function is

$$\mathcal{B}(W) = \theta_0 \frac{(\theta_1 + \max\{W, 0\})^{1-\sigma} - \theta_1^{1-\sigma}}{1 - \sigma},$$

so θ_0 controls the overall strength of the bequest motive and θ_1 controls its wealth dependence. The annual bequest-flow-to-wealth ratio is most informative about θ_0 , while the p90/p50 ratio of total wealth among living PSID households aged 76–84 is most informative about θ_1 . Cross-sectional wealth is measured at the beginning of the model period; bequests are measured after the saving decision at the death node. Together with the aggregate wealth ratio, these moments form a joint three-parameter block for $(\beta, \theta_0, \theta_1)$, following the broader strategy of separating lifecycle saving, bequest intensity, and wealth dispersion in [De Nardi and Yang \(2014\)](#).

1.3 Target Moments and Fit

The target system uses equal squared percentage deviations: the weight on moment m is $1/m_{\text{data}}^2$. This transparent preliminary objective puts the moments in comparable relative units; it is not an estimated optimal GMM weighting matrix. [Table 3](#) reports the complete target system and its strict model counterpart.

Table 3: Target moments and model fit

Moment	Data	Model	Gap	Weight	Loss
<i>Saving and bequests</i>					
Wealth / annual gross labor earnings	6.8731	5.1365	−1.7366	0.0212	0.06384
Annual bequest flow / aggregate wealth	0.0088	0.0081	−0.0007	12913.22	0.00711
Total-wealth p90/p50, ages 76–84	3.4481	1.8926	−1.5556	0.0841	0.20352
<i>Baseline consumption and housing demand</i>					
Childless-renter rent-expenditure slope	0.2647	0.2831	0.0185	14.2734	0.00486
Childless-renter intercept at mean price	0.1064	0.1068	0.0004	88.3061	0.00001
Bottom-quintile childless-renter mean rooms	3.8974	3.8348	−0.0627	0.0658	0.00026
<i>Family size and family housing</i>					
Family-size consumption coefficient	0.0752	0.0752	−0.0000	176.8312	0.00000
Housing increment with the first child, rooms	0.6644	0.6653	0.0009	2.2651	0.00000
Rooms gap, 3+ versus 1–2 children	0.3677	0.3640	−0.0037	7.3963	0.00010
Completed fertility	1.9180	2.0854	0.1674	0.2718	0.00761
Childlessness rate	0.1880	0.1925	0.0045	28.2933	0.00057
<i>Tenure and aggregate housing</i>					
Ownership rate, ages 30–55	0.5755	0.5750	−0.0005	3.0196	0.00000
Four-year residual tenure Brier score	0.1176	0.1177	0.0001	72.2923	0.00000
Mean occupied rooms, ages 18–85	5.7800	5.6876	−0.0924	0.0299	0.00026
Total loss					0.28814

The calibration matches the baseline demand auxiliaries, childlessness, the family housing moments, ownership, tenure dispersion, and aggregate rooms closely. Completed fertility is 2.085, compared with 1.918 in the data. The central failure is the wealth block: the model produces a wealth-to-gross- earnings ratio of 5.136, compared with 6.873, and a late-life wealth p90/p50 ratio of 1.893, compared with 3.448. These two rows account for 92.8 percent of the objective.

The annual discount factor moves to 0.9991, close to its upper search bound of 0.9995, yet the model still underproduces both aggregate wealth and late-life dispersion. A conditional profile is nearly flat at the top: reoptimizing the other parameters at annual discount factors 0.9990 and 0.9995 gives losses 0.2956 and 0.2960. The estimate should therefore not be interpreted as a precise direct estimate of patience. The present specification selects a very patient region because it lacks enough wealth accumulation and dispersion even there.

The calibration remains provisional. In particular, the income-risk process is constrained by Stone–Geary feasibility, the wealth block points to missing late-life heterogeneity, and four parameters are near a search boundary. These limitations should be reported alongside the current fit rather than hidden by a one-to-one identification claim.

References

- Davis, M. A., and S. Van Nieuwerburgh. 2015. “Housing, Finance, and the Macroeconomy.” *Handbook of Regional and Urban Economics* 5: 753–811.
- De Nardi, M., and F. Yang. 2014. “Bequests and Heterogeneity in Retirement Wealth.” *European Economic Review* 72: 182–196.
- Greaney, B., A. Parkhomenko, and S. Van Nieuwerburgh. 2025. “Dynamic Urban Economics.” Working paper.
- Saiz, A. 2010. “The Geographic Determinants of Housing Supply.” *Quarterly Journal of Economics* 125(3): 1253–1296.
- Sommer, K., and P. Sullivan. 2018. “Implications of US Tax Policy for House Prices, Rents, and Homeownership.” *American Economic Review* 108(2): 241–274.